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ORIGINAL PAPER

Novel amylomacins from seaweed-associated *Bacillus amyloliquefaciens* as prospective antimicrobial leads attenuating resistant bacteria

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Abstract
The rise in antibiotic-resistant bacterial strains prompting nosocomial infections drives the search for new bioactive substances of promising antibacterial properties. The surfaces of seaweeds are rich in heterotrophic bacteria with prospective antimicrobial substances. This study aimed to isolate antibacterial leads from a seaweed-associated bacterium. Heterotrophic *Bacillus amyloliquefaciens* MTCC 12716 associated with the seaweed *Hypnea valentiae*, was isolated and screened for antimicrobial properties against drug-resistant pathogens. The bacterial crude extract was purified and three novel amicoumacin-class of isocoumarin analogues, 11′-butyl acetate amicoumacin C (amylomacin A), 4′-hydroxy-11′-methoxyethyl carboxylate amicoumacin C (amylomacin B) and 11′-butyl amicoumacin C (amylomacin C) were isolated to homogeneity. The studied amylomacins possessed potential activities against *Pseudomonas aeruginosa*, vancomycin-resistant *Enterococcus faecalis*, *Klebsiella pneumoniae*, methicillin-resistant *Staphylococcus aureus*, and *Shigella flexneri* with a range of minimum inhibitory concentration values from 0.78 to 3.12 µg/mL, although standard antibiotics ampicillin and chloramphenicol were active at 6.25–25 µg/mL. Noticeably, the amylomacin compound encompassing 4′-hydroxy-11′-methoxyethyl carboxylate amicoumacin C functionality (amylomacin B), displayed considerably greater antagonistic activities against methicillin-resistant *S. aureus*, vancomycin-resistant *E. faecalis*, *Vibrio parahaemolyticus*, *Escherichia coli*, and *K. pneumoniae* (minimum inhibitory concentration 0.78 µg/mL) compared to the positive controls and other amylomacin analogues. Antimicrobial properties of the amylomacins, coupled with the presence of polyketide synthase-I/non-ribosomal peptide synthetase hybrid gene attributed the bacterium as a promising source of antimicrobial compounds with pharmaceutical and biotechnological applications.

Kajal Chakraborty and Vinaya Kizhakkepatt Kizhakkekalam have equally contributed towards the outcome of the paper.

Accession numbers Partial 16S rRNA gene sequence of *B. amyloliquefaciens* was submitted in NCBI GenBank (accession number of KX272634), and functional gene sequence for *pks/nrps* was submitted to the NCBI gene bank with an accession number of MH157092.

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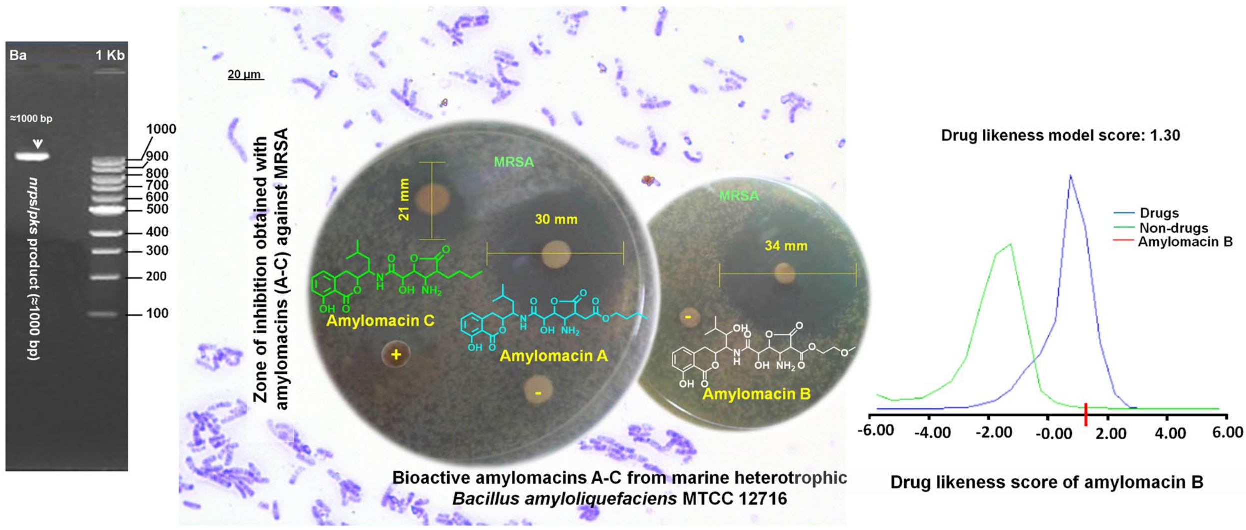
Graphic abstract



Keywords *Bacillus amyloliquefaciens* · Amylomacins · Antimicrobial · Type-I polyketide synthase/non ribosomal peptide synthetase hybrid gene

Introduction

Multidrug-resistant pathogens, such as vancomycin-resistant *Enterococcus faecalis* (VRE*fs*) and methicillin-resistant *Staphylococcus aureus* (MRSA) were described to cause nosocomial infections (Amalaradjou and Venkitanarayanan 2014). The immune-compromised patients and those treated under intensive care are at greater likelihood for co-colonization with these endemic strains causing increased severity of illness. Rise in antibiotic-resistant bacterial strains prompting nosocomial infections and the necessity for antimicrobial agents drives the search for bioactive substances of significantly greater antimicrobial properties (Meklat et al. 2011).

Heterotrophic bacterial flora associated with the seaweeds were found to biosynthesize specialized antimicrobial compounds that inhibit the competitive pathogenic microorganisms (Bernan et al. 1997). More than four hundred strains of surface-associated bacteria from seaweeds and invertebrates were isolated from the coastal waters of Scotland, and greater than 30% of those were found to be prospective antimicrobial agents (Burgess et al. 1999). More than two hundred bacteria were characterized for production of antibiotic compounds from seaweed-associated heterotrophs (Lemos et al. 1985). Besides, the fraction of active bacteria associated with the seaweeds was higher (> 10%) than those obtained from sediment (~ 5%) and seawater (~ 7%) (Zheng et al. 2005). Heterotrophic bacterial species associated with the seaweed *Jania rubens* collected from the Mediterranean Sea were isolated, where 36% of the isolates were found to produce antibiotic-like agents with prospective inhibition against pathogenic bacteria and fungi (Ismail-Ben et al. 2012). O-heterocyclic derivatives with antibacterial properties were isolated from marine bacterium *Bacillus subtilis* associated with seaweed (Chakraborty et al. 2017a). Antibacterial furanoterpenoids and a macrolactin analogue were previously characterized from the seaweed-associated symbiont *Bacillus subtilis* MTCC 10403 (Chakraborty et al. 2014, 2017b). Antibacterial polyketides and macrocyclic lactones were characterized from *B. amyloliquefaciens* dwelling in association with the seaweeds *Laurenciae papillosa*, *Padina gymnospora*, and *Hypnea valentiae* (Chakraborty et al. 2017c, 2018, 2020).

The heterogeneity of seaweeds and the accompanying heterotrophic bacterial symbionts comprised valuable sources of antimicrobial agents with prospective pharmaceutical applications (Thilakan et al. 2016). Antibacterial activity analysis integrated with the presence of polyketide synthase/non-ribosomal peptide synthetase (*pks*/*nrps*) genes coding for the bioactive compounds displayed that seaweed-associated bacteria had large-spectrum of antibacterial activity (Andryukov et al. 2019). The seaweed-associated *Bacillus* belonging to *Firmicutes* was found to be prospective source of antibacterial compounds, for example polyketides, non-ribosomal cyclotides, and hybrids of polyketide-peptide (e.g., amicoumacins) (Li et al. 2015). *Bacillus* sp. were recognized to produce a considerable proportion of bioactive metabolites, which were reported to be coded by genes exemplifying up to 8% of the genome (Chen et al. 2017; Kim et al. 2012).

Previous literature reports revealed that the seaweed-associated bacteria belonging to the phylum *Firmicutes* were potential sources of bioactive compounds with polyketide frameworks (Thilakan et al. 2016; Chakraborty et al. 2017a; Kizhakkekalam and Chakraborty 2019; Kizhakkekalam et al. 2020). The organic extract of *B. amyloliquefaciens* MTCC 12716 associated with seaweed *H. valentiae* (Turner) Montagne (family *Cystocloniaceae*) of the southeast coast of Indian peninsular was chosen for purification and chemical investigation, based on its significant antibacterial activities (Kizhakkekalam and Chakraborty 2019). In this background, the present study envisages the characterization of novel isocoumarin-type of homologous amicoumacins, named as amylomacins A–C from the studied bacterium (Fig. 1), and evaluating them against multidrug-resistant pathogens. The potential of *B. amyloliquefaciens* MTCC 12716 to produce amicoumacin metabolites were further verified by polymerase chain reaction by the amplification of gene encoding for polyketide synthase (*pks*) and non-ribosomal peptide synthetase (*nrps*) hybrid. A *pks-nrps* gene-encrypted biosynthetic pathway was hypothesized in analogy with the amplified hybrid gene.

Materials and methods

Isolation and antimicrobial screening of heterotrophic *B. amyloliquefaciens*

Heterotrophic bacteria living in association with *H. valentiae*, which were collected from the Gulf of Mannar of the Indian peninsular (9° 17′ 0″ N, 79° 7′ 0″ E) of India, were purified by following previous reports of literature (Thilakan et al. 2016). The heterotrophic bacterial isolates were evaluated for antibacterial properties against the clinically significant pathogens by spot-over-lawn experiment (Kizhakkekalam and Chakraborty 2019). Selected bioactive strain *B. amyloliquefaciens* MTCC 12716 with greater antimicrobial properties against these pathogens was identified by combined phenotypic and genotypic methodologies (Kizhakkekalam and Chakraborty 2020). The pathogenic test organisms, such as vancomycin-resistant *Enterococcus faecalis* (VRE*fs*) ATCC 51299, *Plesiomonas shigelloides* ATCC 14029, *Pseudomonas aeruginosa* ATCC 27853, *Klebsiella pneumoniae* ATCC 13883, and methicillin-resistant *Staphylococcus aureus* (MRSA) ATCC 33592 were procured from the American Type Culture Collection (Manassas, VA, USA). *Yersinia enterocolitica* MTCC 859, *Vibrio parahaemolyticus* MTCC 451, *Shigella flexneri* MTCC 1457, and *Escherichia coli* MTCC 443 were procured from the Microbial Type Culture Collection and Gene Bank (MTCC) of the Institute of Microbial Technology (Chandigarh, India).

Cultivation and extraction of bacterial metabolites

The seaweed-associated *B. amyloliquefaciens* was grown in a modified medium of basal salt agar containing glucose, peptone, potassium dihydrogen phosphate (KH_2PO_4), ammonium nitrate (NH_4NO_3), sodium succinate, along with chloride salts of sodium (NaCl), potassium (KCl), magnesium ($MgCl_2$), and calcium ($CaCl_2$) before being incubated at 37 °C. Ethyl acetate (EtOAc) was used to extract the culture medium to yield the organic extract (5.1 g) (Kizhakkekalam and Chakraborty 2019). The crude (~5 g) was fractionated by flash chromatography over fine silica gel (230–400 mesh)

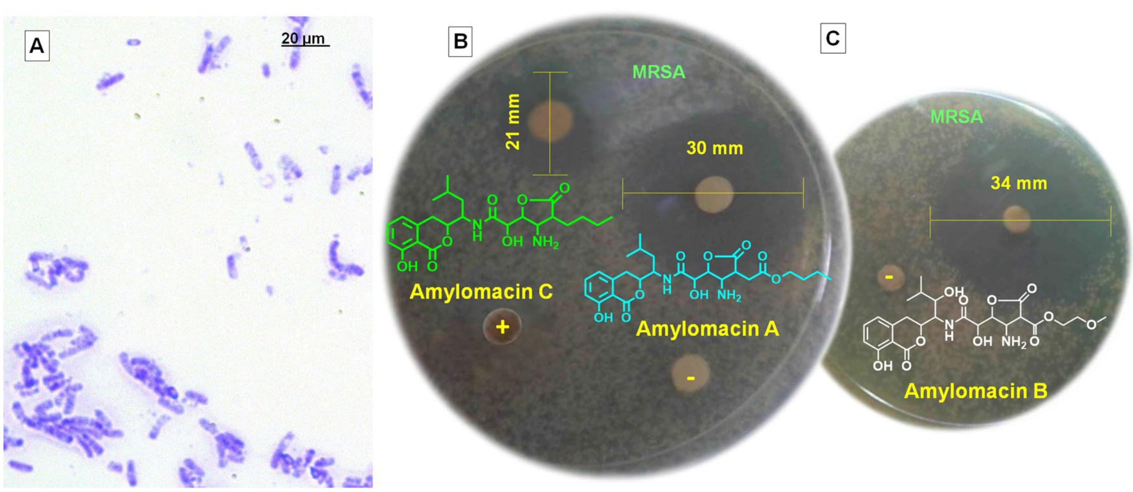


Fig. 1 **A** Gram staining photomicrograph of *B. amyloliquefaciens* MTCC 12716. **B** Zone of inhibition obtained with the amicoumacins (amylomacin A and C). **C** Antibacterial inhibition zone (diameter of the clear zone) exhibited by amylomacin B against methicillin-resistant *S. aureus*

with a series of solvent systems in an increasing polarity (*n*-hexane/EtOAc to methanol MeOH), and the collection of the fractions were monitored at the wavelength (λ) of 248 nm. The fractions acquired from the column were combined into five (B_1–B_5) after TLC (*n*-hexane:EtOAc, 3:1, v/v) and HPLC {on a C_{18} reverse-phase (RP) stationary phase, MeOH:H_2O, 1:1 v/v, isocratic} analyses, before being re-fractionated through the preparatory HPLC (H_2O:MeOH, 1:1 v/v, isocratic). The antibacterial fractions (10 mL each) obtained from B_4 were latter combined to three (B_{4-1} through B_{4-3}), and their homogeneity was ascertained based on TLC (*n*-hexane:EtOAc, 9:1 v/v) and HPLC (fractionated on a C_{18} RP stationary phase, H_2O:MeOH 1:1 v/v, isocratic) analyses. The fraction B_{4-1} (68 mg, light brown oil), B_{4-2} (92 mg, yellowish oil) and B_{4-3} (83 mg, colorless oil) could result in homogenous isocoumarin-type of polyketide-spanned amicoumacin derivatives, named as amylomacins A through C, respectively, with greater than 98% purity.

Spectroscopic analysis

Amylomacin A [11′-butyl acetate amicoumacin C] {butyl 14′-(10′-amino-9′-(8′-hydroxy-7′-((5′-(8-hydroxy-1-oxoisochroman-3-yl)-3′-methylbutyl) amino)-7′-oxoethyl)-12′-oxotetrahydrofuran-11′-yl) acetate}

Yellowish oil; UV $\lambda_{max\ MeOH}$ (log ε): 247 nm (3.42), 315 (3.24); $[\alpha]^{27}{}_{D}$ − 12.3° (MeOH, $c = 21$); TLC (silica GF_{254}; EtOAc:*n*-hexane, 1:9 v/v) R_f: 0.52; R_t (C_{18}-RP HPLC; C_{18} 25 cm × 460 mm, 5 µm; H_2O:MeOH, 1:1 v/v): 2.71 min (Fig. S1); 1D/2D NMR (Table 1; Figs. S2–S8); GC–MS (EI-MS): *m/z* found 520.2 $[M]^+$ for $C_{26}H_{36}N_2O_9$; HRESIMS: *m/z* found 521.2502 $[M+H]^+$, calcd. for $C_{26}H_{37}N_2O_9$ 521.2498 ($\Delta = 0.8$ ppm) (Figs. S9–S11); FT-IR (ν stretching, δ bending, ρ rocking) (cm^{-1}): 3435 (O-H_ν), 3232 (N-H_ν), 2961, 2873 (C-H_ν), 1732 (ester C=O_ν), 1639 (amide C=O_ν), 1506 (aromatic C=C_ν), 1352 (C-H_δ), 1115 (C-O-C_ν), 1008 (hydroxyl C-O_ν), 922, 747 (C-H_δ) (Fig. S12).

Amylomacin B [4′-hydroxy-11′-methoxyethyl carboxylate amicoumacin C] {15′-methoxyethyl 10′-amino-9′-(8′-hydroxy-7′-((4′-hydroxy-5′-(8-hydroxy-1-oxoisochroman-3-yl)-3′-methylbutyl)amino)-7′-oxoethyl)-12′-oxotetrahydrofuran-13′-carboxylate}

Colorless oil; UV $\lambda_{max\ MeOH}$ (log ε): 247 nm (3.41), 310 (3.25); $[\alpha]^{27}{}_{D}$ − 12.1° (MeOH, $c = 21$); TLC (GF_{254}; EtOAc:*n*-hexane, 1:9 v/v) R_f: 0.55; R_t (C_{18}-RP HPLC, H_2O:MeOH, 1:1 v/v): 2.81 min (Fig. S13); 1D/2D NMR (Table 1; Figs. S14–S20); GC–MS (EI-MS): *m/z* found 524.2 $[M]^+$ for $C_{24}H_{32}N_2O_{11}$; HRESIMS: *m/z* found 525.2088 $[M+H]^+$, calcd. for $C_{24}H_{33}N_2O_{11}$ 525.2084 ($\Delta = 0.8$ ppm) (Fig. S21-S23); FT-IR (cm^{-1}): 3404 (O-H_ν), 3225 (N-H_ν), 2946 (C-H_ν), 1737 (ester C=O_ν), 1637 (amide C=O_ν), 1476 (aromatic C=C_ν), 1316 (C-H_δ), 1174 (C-O-C_ν), 1008 (hydroxyl C-O_ν), 965, 727 (C-H_δ) (Fig. S24).

Amylomacin C [11′-butyl amicoumacin C] {8′-(10-amino-11′-butyl-12′-oxotetrahydrofuran-9′-yl)-8′-hydroxy-*N*-(7′-(8-hydroxy-1-oxoisochroman-3-yl)-3′-methylbutyl) acetamide}

Colorless oil; UV $\lambda_{max\ MeOH}$ (log ε): 249 nm (3.40), 312 (3.26); $[\alpha]^{27}{}_{D}$ − 12.6° (MeOH, $c = 21$); TLC (GF_{254}; EtOAc:*n*-hexane, 1:9 v/v) R_f: 0.56; R_t (C_{18}-RP HPLC, H_2O:MeOH, 1:1 v/v): 2.90 min (Fig. S25); 1D/2D NMR (Table 1; Figs. S26–S32); GC–MS (EI-MS): *m/z* found 462.2 $[M]^+$ for $C_{24}H_{34}N_2O_7$; HRESIMS: *m/z* found 463.2448 $[M+H]^+$, calcd. for $C_{24}H_{35}N_2O_7$ 463.2440 ($\Delta = 1.7$ ppm) (Figs. S33–S35); FT-IR (cm^{-1}): 3434 (O-H_ν), 3235 (N-H_ν), 2955 (C-H_ν), 1736 (ester C=O_ν), 1636 (amide C=O_ν), 1466 (aromatic C=C_ν), 1312 (C-H_δ), 1118 (C-O-C_ν), 1074 (hydroxyl C-O_ν), 965, 757 (C-H_δ) (Fig. S36).

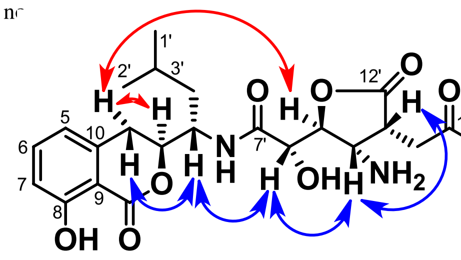
Antibacterial activities

The purified amylomacins A–C were assessed for antimicrobial potential by microdilution and disc diffusion methods, as reported earlier (Kizhakkekalam and Chakraborty 2020) (Supplementary material S1). Antibacterial activities were expressed as minimum inhibitory concentration (MIC, µg/mL) and zone of inhibition (mm) against pathogenic bacteria. Minimum inhibitory concentration (MIC) was performed by microdilution method with {3-(4,5-dimethylthiazol-2-yl) 2,5-diphenyltetrazolium bromide}-formazan (MTT) colorimetric endpoint detection on Mueller Hinton (MH) agar. The live cells were documented as purple color by the formation of formazan crystals. Bactericidal zones of the active compounds were envisaged as yellow color in the plate. Solvent ethyl acetate and commercially available antibiotic disc (HiMedia Laboratories, LLC, Kelton, PA) were used as negative and positive control, respectively.

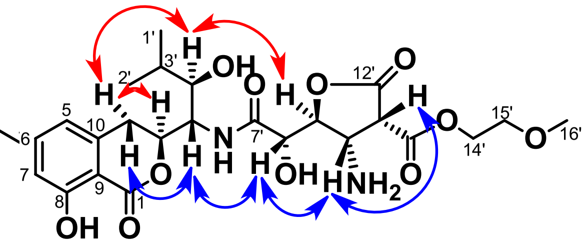
Amplification of gene for amylomacin biosynthesis

Genomic DNA was isolated from *B. amyloliquefaciens*, and further analyzed for the presence of biosynthetic genes, with specific primers "GCNGGYGGYGCNTAYGTNCC" (forward) and "CCNCGDATYTTNACYTG" (reverse), to ascertain the potential of the studied bacterium to produce the antimicrobial metabolites of polyketide/polypeptide classes. PCR amplification was performed by a reported method (Kizhakkekalam and Chakraborty 2020; Zhao et al. 2008). The expected product of ≈ 1 kb was purified with Gel Elute™ (Sigma-Aldrich, St. Louis, MO), and was sequenced before being submitted in the NCBI GenBank.

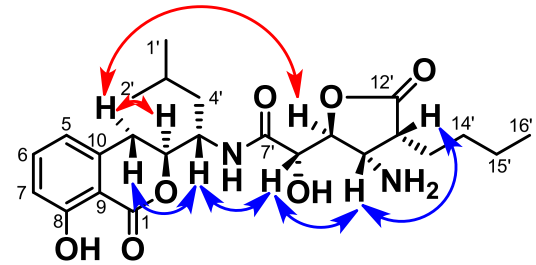
Table 1 NMR spectroscopic data of amylomacins A–C



Amylomacin A



Amylomacin B

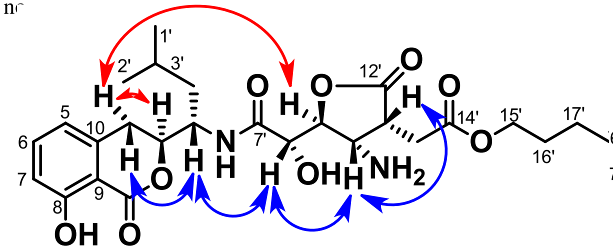


Amylomacin C

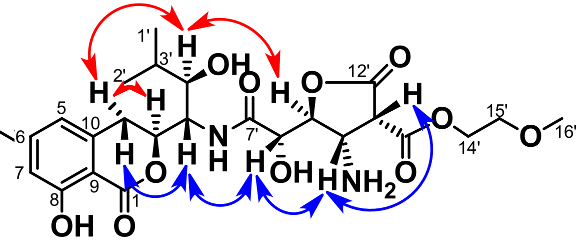
C. no	Amylomacin A				Amylomacin B				Amylomacin C			
	^{13}C	^{1}H (int., mult., J in Hz)‡	^{1}H-^{1}H COSY	HMBC	^{13}C	^{1}H (int., mult., J in Hz)‡	^{1}H-^{1}H COSY	HMBC	^{13}C	^{1}H (int., mult., J in Hz)‡	^{1}H-^{1}H COSY	HMBC
1	166.3	–	–	–	166.2	–	–	–	167.1	–	–	–
2	–	–	–	–	–	–	–	–	–	–	–	–
3	71.8	4.57 (1Hα, m)	H-4, H-5′	C-1,9	72.6	4.60 (1Hα, m)	H-4, H-5′	C-1,9	71.8	4.60 (1Hα, m)	H-4, H-5′	C-1,9
4	38.6	2.89 (1Hα, d, 2.5) 3.07 (1Hβ, d, 2.9)	–	C-10	38.6	2.81 (1Hα, d, 2.1) 3.10 (1Hβ, d, 2.3)	–	C-10,5′	38.9	2.89 (1Hα, d, 2.9) 3.12 (1Hβ d, 3.2)	–	C-10
5	128.6	7.20 (1H, d, 6.7)	H-6	C-4,10	128.6	7.26 (1H, d, 6.4)	H-6	C-4,10	128.9	7.18 (1H, d, 6.4)	H-6	C-4,10
6	129.2	7.33 (1H, t, 6.5)	H-7	C-5	129.3	7.33 (1H, t, 6.6)	H-7	C-5	130.9	7.30 (1H, t, 6.1)	H-7	C-5
7	115.3	6.47 (1H, d, 6.1)	–	C-8,9	116.1	6.10 (1H, d, 6.0)	–	C-8,9	115.9	6.61 (1H, d, 6.8)	–	C-8,9
8	155.9	–	–	–	154.4	–	–	–	153.1	–	–	–
9	116.1	–	–	–	117.2	–	–	–	112.1	–	–	–
10	135.9	–	–	–	130.3	–	–	–	133.0	–	–	–
1′	23.3	1.00 (3H, d, 3.0)	H-3′	C-3′,4′	21.2	1.01 (3H, d, 3.1)	H-3′	C-4′	21.6	0.99 (3H, d, 4.2)	H-3′	C-3′,4′
2′	22.7	0.96 (3H, d, 2.8)	H-3′	C-3′,4′	22.7	0.98 (3H, d, 2.9)	H-3′	C-3′,4′	22.3	0.96 (3H, d, 4.1)	H-3′	C-3′,4′
3′	37.4	2.05 (1H, m)	H-4′	–	38.5	2.08 (1H, m)	H-4′	–	37.6	2.02 (1H, m)	H-4′	–
4′	42.6	1.92 (2H, t, 3.1)	H-5′	–	71.2	3.89 (1Hα, t, 4.3)	H-5′	C-5′,3′	40.1	1.96 (2H, t, 3.2)	H-5′	–
5′	45.5	4.25 (1Hβ, dt, 3.1, 2.0)	–	C-3,4,4′,7′	45.5	4.31 (1Hβ, t, 4.2)	–	C-3,7′	46.7	4.33 (1Hβ, dt, 4.4, 2.1)	–	C-3,4,4′,7′
6′	–NH–	7.67 (1H, d, 3.0)	–	C-5′,7′	–NH–	7.65 (1H, d, 3.1)	–	C-5′,7′	–NH–	7.63 (1H, d, 3.4)	–	C-5′,7′
7′	169.5	–	–	–	170.4	–	–	–	171.3	–	–	–
8′	75.0	4.32 (1Hβ, d, 3.1)	H-9′	C-7′,9′	72.2	4.50 (1Hβ, d, 4.8)	H-9′	C-7′,9′	73.2	4.41 (1Hβ, d, 4.2)	H-9′	C-7′,9′
9′	81.3	4.47 (1Hα, t, 4.2)	H-10′	C-7′	80.7	4.14 (1Hα, t, 5.1)	H-10′	C-7′	83.5	4.51 (1Hα, d, 4.2)	H-10′	C-7′
10′	36.8	3.56 (1Hβ, t, 2.4)	H-11′	C-9′,8′,11′,12′	36.8	3.57 (1Hβ, t, 3.2)	H-11′	C-8′,9′,11′,12′,13′	38.5	3.41 (1Hβ, d, 4.2)	H-11′	C-8′,9′,11′,12′
–	NH_2	6.87 (2H, d, 2.5)	–	C-10′	NH_2	6.89 (2H, d, 2.7)	–	C-10′	NH_2	6.83 (2H, d, 2.6)	–	C-10′

Table 1 (continued)

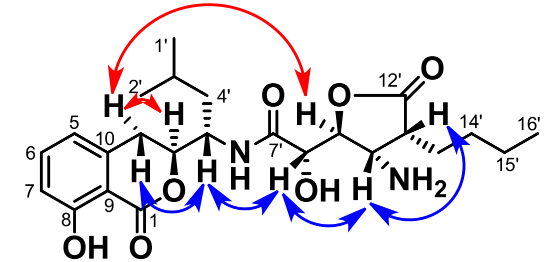
C. n°



Amylomacin A



Amylomacin B



Amylomacin C

	Amylomacin A				Amylomacin B				Amylomacin C			
	^{13}C	^{1}H (int., mult., *J* in Hz)‡	^{1}H-^{1}H COSY	HMBC	^{13}C	^{1}H (int., mult., *J* in Hz)‡	^{1}H-^{1}H COSY	HMBC	^{13}C	^{1}H (int., mult., *J* in Hz)‡	^{1}H-^{1}H COSY	HMBC
11′	68.4	3.15 (1H*β*, dt, 2.4, 1.8)	H-13′	C-9′,14′	59.0	3.14 (1H*β*, d, 2.6)	–	C-9′,13′	41.8	3.01 (1H*β*, d, 4.2)	H-13′	C-9′,14′
12′	170.8	–	–	–	170.6	–	–	–	170.3	–	–	–
13′	38.5	2.63 (2H, d, 4.8)	–	C-14′	169.5	–	–	–	30.6	2.32 (2H, m)	H-14′	–
14′	170.5	–	–	–	68.5	4.09 (2H, t, 2.8)	H-15′	C-13′,15′	24.8	1.72 (2H, m)	H-15′	–
15′	65.6	4.11 (2H, t, 3.2)	H-16′	C-14′,17′	73.7	3.66 (2H, t, 3.5)	–	C-14′,16′	22.7	1.29 (2H, m)	H-16′	C-16′
16′	31.4	1.53 (2H, m)	H-17′	C-17′	53.4	3.34 (3H, s)	–	C-15′	14.1	0.88 (3H, t, 7.6)	–	C-14′
17′	21.2	1.27 (2H, m)	H-18′	–	–	–	–	–	–	–	–	–
18′	19.2	1.08 (3H, t, 6.9)	–	C-17′	–	–	–	–	–	–	–	–

Assignments were made with the aid of COSY, HSQC, HMBC and NOESY experiments. NOESY relations were represented by double-sided arrows, wherein the blue and red-colored arrows indicated beta and alpha protons, respectively

‡NMR spectra were recorded using Bruker AVANCE III 500 MHz (AV 500) spectrometer in $CDCl_3$ as aprotic solvent at ambient temperature with TMS as the internal standard (δ 0 ppm). Values in parts per million (ppm), multiplicity and coupling constants ($J = $Hz) are indicated in parentheses

Statistical analysis

Statistical Program for Social Sciences 13.0 (SPSS Inc, Chicago, IL, ver. 10.0) with one-way analysis of variance calculated the significant differences between means and were labeled as $p < 0.05$. The results were depicted as mean of triplicates along with standard deviation.

Results

Characterization of seaweed-associated *B. amyloliquefaciens* MTCC 12716

The heterotrophic bacterium purified from the rhodophytan seaweed was identified by inclusive genotypic and phenotypic characterization, as Gram positive, with white rippled and lobed edges, possessing nitrate reduction and casein hydrolyzing abilities with positive amylase activity. Further, the MALDI-TOF confidence count of 1.953 and 99% BLAST likeness search recognized the bacterium as *B. amyloliquefaciens* belonging to the class of *Firmicutes*. The obtained sequence of 16S rRNA of the studied *B. amyloliquefaciens* was deposited in the NCBI Genbank (acquisition number of KX272634), and the bacterial strain was submitted as microbial type culture collection (MTCC 12716) in IMTECH (Institute of Microbial Technology, Chandigarh of India).

Structural characterization of amylomacins A-C

Amylomacin A (11′-butyl acetate amicoumacin C) was purified to homogeneity by repeated chromatographic experiments including preparatory high-performance liquid chromatography (HPLC) (Fig. S1). The molecular formula of amylomacin A was deduced as $C_{26}H_{36}N_2O_9$ by mass spectroscopic {HRESIMS: *m/z* found 521.2502 $[M+H]^+$, calcd. for $C_{26}H_{37}N_2O_9$ 521.2498}, and NMR/IR experiments (Table 1; Figs. 2A, S2–S12; Structure S1).

Amylomacin B (4′-hydroxy-11′-methoxyethyl carboxylate amicoumacin C) was purified to homogeneity by preparatory HPLC (Fig. S13), and its empirical formula ($C_{24}H_{32}N_2O_{11}$) and structure were elucidated by mass spectroscopy {HRESIMS: *m/z* found 525.2088 $[M+H]^+$, calcd. for $C_{24}H_{33}N_2O_{11}$ 525.2084} and NMR/IR (Table 1; Figs. 2B, S14–S24; Structure S2). Amylomacin C (11′-butyl amicoumacin C) was purified to homogeneity by repeated chromatographic experiments including preparatory HPLC (Fig. S25). The molecular formula of amylomacin C was deduced as $C_{24}H_{34}N_2O_7$ by mass spectroscopic {HRESIMS: *m/z* found 463.2448 $[M+H]^+$, calcd. for $C_{24}H_{35}N_2O_7$ 463.2440} and NMR/IR experiments (Table 1; Figs. 2C, S26–S36; Structure S3).

Antimicrobial properties of amylomacins

The heterotrophic *B. amyloliquefaciens* was found to possess a large spectrum of antimicrobial activity against MRSA

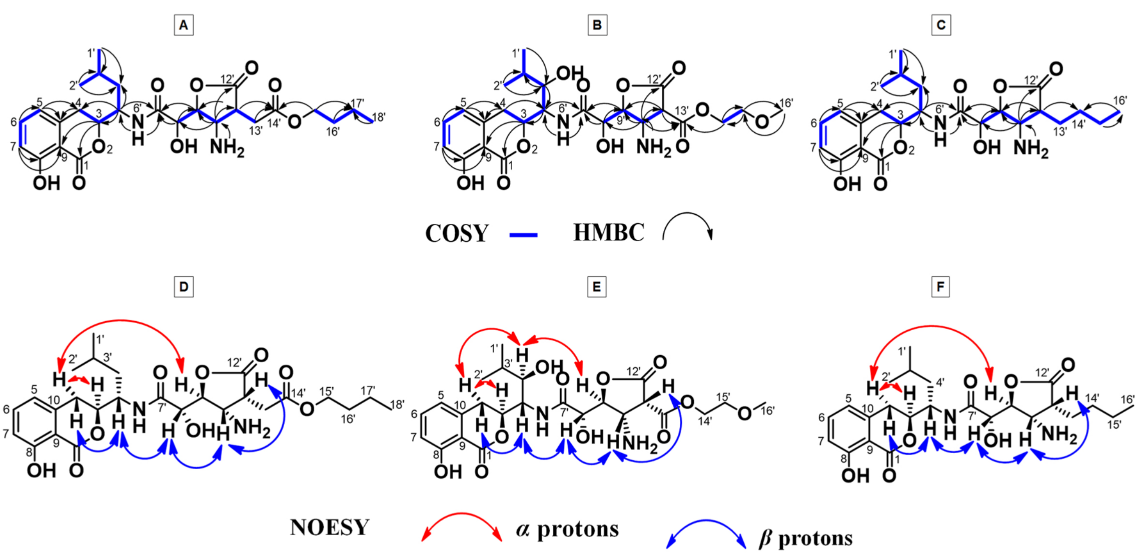


Fig. 2 A–C ^{1}H-^{1}H COSY/HMBC correlations of 11′-butyl acetate amicoumacin C (amylomacin A), 4′-hydroxy-11′-methoxyethyl carboxylate amicoumacin C (amylomacin B) and 11′-butyl amicoumacin C (amylomacin C). The key ^{1}H-^{1}H COSY couplings were characterized with bold-faced bonds. HMBC couplings were exemplified by double-barbed arrows. **D–F** NOESY relationships (designated as double-sided arrows, in which the blue and red-colored ones pointed to *β*- and *α*-protons, respectively (of amylomacins A, B, and C, respectively). Spectroscopic analyses encompassing one and two-dimensional NMR were documented with a NMR spectrometer (Bruker Avance DPX; 500 MHz) using deuteriated chloroform as a solvent

(16 mm), VRE*fs* (14 mm), drug-resistant *P. aeruginosa* (15 mm) and *K. pneumoniae* (18 mm), along with other clinical pathogens as evaluated by spot-on-lawn assay. Amylomacins A–C showed significant antibacterial properties against all the tested strains, amongst that amylomacin B exhibited promising inhibitory potential against MRSA (34 mm), VRE*fs* (29 mm), *P. aeruginosa* (27 mm) and *K. pneumoniae* (30 mm) (30 µg on disc). The lowest concentration of amylomacins that inhibited the visible growth of pathogens used in the test following overnight incubation were noted as MIC values (0.78–0.6.25 µg/mL) (Table 2, Fig. 1). Conspicuously, the antibiotic agents ampicillin and chloramphenicol (30 µg) exhibited insignificant inhibition zone.

Microdilution analysis showed considerable MICs with 4′-hydroxy-11′-methoxyethyl carboxylate amicoumacin C (amylomacin B) for MRSA, VRE*fs*, *K. pneumoniae*, *V. parahaemolyticus*, *E. coli*, and *P. aeruginosa* (0.78 µg/mL) compared to the positive controls (chloramphenicol and ampicillin with a range of MIC values from 6.25 to 25 µg/mL) and other amylomacin analogues (MIC of 1.56–3.12 µg/mL). These results could recognize the greater antibacterial property of amylomacin B when compared to the commercial standards and other studied compounds.

Discussion

Multidrug-resistant (MDR) bacteria have often been described as a major concern to public health and are associated with nosocomial infections. The clinical isolates, such as VRE*fs*, MRSA, drug-resistant *K. pneumoniae* and *P. aeruginosa* were found to develop antibiotic resistance and spread in the hospital environment. *Shigella flexneri* is a highly virulent pathogen, which causes the most communicable of bacterial dysenteries, shigellosis (Zaidi and Estrada-García 2014). Earlier studies demonstrated that the seaweed-associated heterotrophs could possess significantly higher biological activities than the free-living microbial flora in the ocean (Bengtsson et al. 2010). They biosynthesize specialized bioactive metabolites of *pks* or *nrps* origin, and were reported as prospective antimicrobial agents to inhibit human pathogens (Ismail-Ben et al. 2012). Among various specialized metabolites characterized from seaweed-associated bacteria, amicoumacins belonging to a larger group of dihydroisocoumarin natural products derived from the *pks*/*nrps* hybrid-derived clusters (Fig. 3), were identified in the marine *Bacillus* sp. with potential pharmacological activities (Bai et al. 2014; Komaki et al. 2016).

The structural properties of the isolated amylomacin A were comparable with the previously reported dihydroisocoumarin-derived amicoumacins, which were found to be δC 165–170 (C-1), δH 4.55–4.70/δC 70–85 (C-3), δH 2.80–2.90, 3.00–3.10/δC 30–40 (C-4), δH 6.8–7.20/δC 119–130 (C-5), δH 7.3–7.5/δC 129–137 (C-6), δH 6.4–6.9/δC 113–117 (C-7), δC 155–161 (C-8), δC 108–117 (C-9), δC 135–142 (C-10) (Bai et al. 2014; Tang et al. 2016). The first substructure of amylomacin A was deduced as isochromanone through COSYs from H-5 to H-7 and H-3 to H-4 along with HMBCs from H-4 to C-10, H-3 to C-1, C-9. The deshielded carbon/proton nuclei were suggestive

Table 2 Antibacterial activity of amylomacins A–C and commercial standards against clinical pathogens

Antibacterial activity	Zone of inhibition (mm)				
	Amylomacin A	Amylomacin B	Amylomacin C	Chloramphenicol	Ampicillin
MRSA ATCC 33592	$30.00^{a} \pm 0.01$ (1.56)	$34.00^{a} \pm 0.01$ (0.78)	$21.00^{b} \pm 0.05$ (3.12)	$14.00^{c} \pm 0.02$ (6.25)	$10.00^{c} \pm 0.02$ (12.50)
VRE*fs* ATCC 51299	$25.00^{b} \pm 0.04$ (3.12)	$29.00^{a} \pm 0.04$ (0.78)	$20.00^{b} \pm 0.02$ (3.12)	$13.00^{c} \pm 0.01$ (6.25)	$13.00^{c} \pm 0.04$ (25.00)
P. aeruginosa ATCC 27853	$22.00^{b} \pm 0.02$ (1.56)	$27.00^{a} \pm 0.02$ (1.56)	$19.00^{b} \pm 0.03$ (3.12)	$11.00^{c} \pm 0.02$ (12.50)	$7.00^{c} \pm 0.01$ (12.50)
K. pneumoniae ATCC 13883	$27.00^{a} \pm 0.02$ (1.56)	$30.00^{a} \pm 0.02$ (0.78)	$21.00^{b} \pm 0.03$ (3.12)	$7.00^{c} \pm 0.04$ (12.50)	$7.00^{c} \pm 0.02$ (12.50)
S. flexneri MTCC 1457	$24.00^{b} \pm 0.03$ (3.12)	$28.00^{a} \pm 0.05$ (3.12)	$22.00^{b} \pm 0.04$ (3.12)	$14.00^{c} \pm 0.01$ (6.25)	$16.00^{c} \pm 0.06$ (6.25)
E. coli MTCC 443	$23.00^{b} \pm 0.06$ (1.56)	$30.00^{a} \pm 0.01$ (0.78)	$19.00^{b} \pm 0.03$ (6.25)	$16.00^{c} \pm 0.02$ (6.25)	$12.00^{c} \pm 0.01$ (12.50)
P. shigelloides ATCC 14029	$24.00^{b} \pm 0.01$ (3.12)	$29.00^{a} \pm 0.02$ (1.56)	$22.00^{b} \pm 0.01$ (6.25)	$15.00^{c} \pm 0.03$ (6.25)	$13.00^{c} \pm 0.05$ (6.25)
Y. enterocolitica MTCC 859	$21.00^{b} \pm 0.01$ (6.25)	$21.00^{b} \pm 0.01$ (3.12)	$21.00^{b} \pm 0.01$ (3.12)	$11.00^{c} \pm 0.01$ (6.25)	$15.00^{c} \pm 0.01$ (6.25)
V. parahaemolyticus MTCC 451	$23.00^{b} \pm 0.05$ (1.56)	$29.00^{a} \pm 0.04$ (0.78)	$24.00^{b} \pm 0.04$ (3.12)	$15.00^{c} \pm 0.06$ (6.25)	$16.00^{c} \pm 0.02$ (6.25)

The heterotrophic bacterium *B. amyloliquefaciens* MTCC 12716 cultured in the modified basal salt agar media composed of peptone (5.0 g), KH_2PO_4 (0.05 g), $MgCl_2{\cdot}H_2O$ (0.2 g), FeEDTA (0.005 g), $CaCl_2{\cdot}2H_2O$ (0.02 g), KCl (0.7 g), NH_4NO_3 (1.0 g), Tris (1.0 g), sodium succinate (5.0 g), glucose (1.35 g) and NaCl (25 g)

The antibacterial activity in terms of minimum inhibitory concentration in µg/mL and inhibitory zone against clinically relevant pathogens in mm

MRSA methicillin-resistant *S. aureus*, VRE*fs* vancomycin-resistant *E. feacalis*

[a–c]Row-wise values with different superscripts (a–c) of this type indicate significant difference ($p < 0.05$), which implied for the statistical evaluation of the results. Triplicate values were taken and the variance analyses and results were expressed as mean ± SD (n = 3)

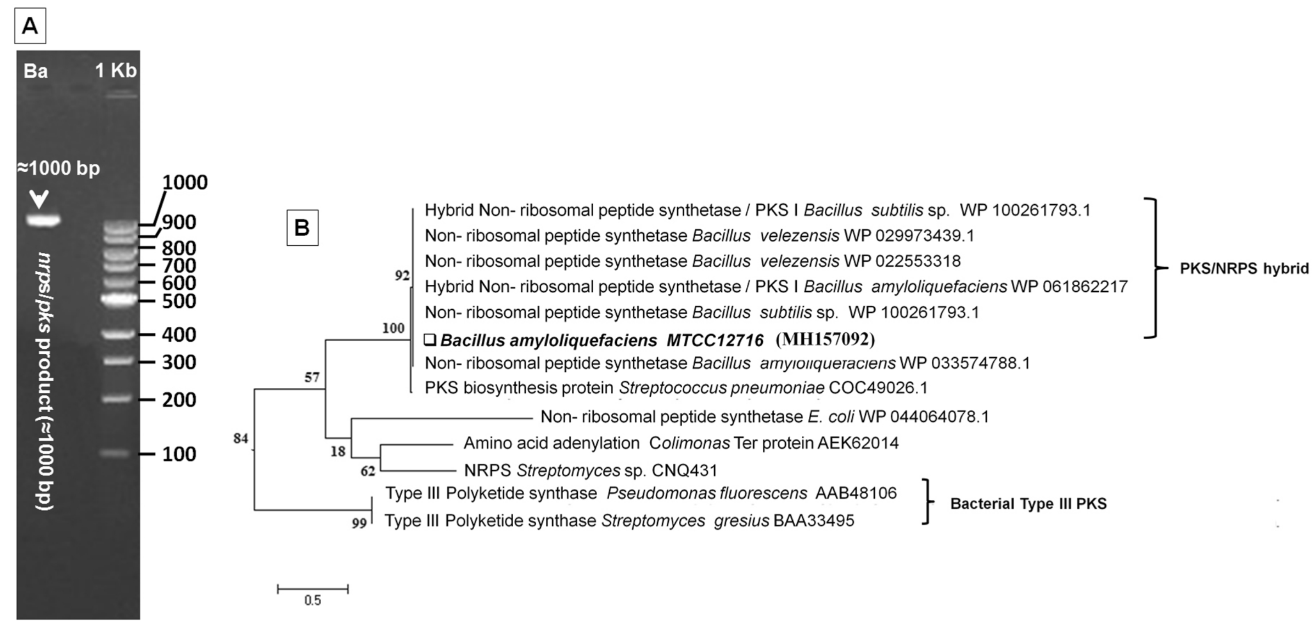


Fig. 3 **A** Bacterial strain showed positive hit for type-I polyketide synthases (*pks*-I)/non-ribosomal peptide synthetases (*nrps*) hybrid gene product (≈ 1000 bp) (lane 1) of *B. amyloliquefaciens* MTCC 12716 (Ba) and molecular marker (lane 2). PCR conditions were set as initial denaturation at 94 °C for a time of 5 min followed by 30 cycles of 30 s at 94 °C, 30 s at 56 °C, 1 min at 72 °C, and a final extension of 5 min at 72 °C. The amplification was accomplished in a total reaction volume of 25 µL having 10× buffer with $MgCl_2$ (Sigma), 0.25 mM dNTPs (Sigma), 1.0 mM of each primer (Sigma), 1 ng DNA, and 0.3 U Taq polymerase (Sigma). Agarose gel (1.5%) was used to perceive the amplified products as fragments, by electrophoresis. The amplified gene sequence for type-I *pks/nrps* was submitted to the NCBI gene bank (accession identity of MH157092). **B** Molecular phylogeny analysis of the hybrid of *nrps/pks*. Evolutionary history was deduced by maximum likelihood method (JTT matrix-centered model). An aggregate of 152 positions were located in the final dataset. Evolutionary analysis was effected in MEGA6

of ester groups among C-1 and C-3, which deduced the isochromanone (dihydroisocoumarin) part. The shift at δ 7.67 was represented by –NH– proton and its position at H-6′ was confirmed through HMBCs from δ 7.67 (–NH–) to C-7′, C-5′, and H-5′ to C-7′. These data were comparable with previous amicoumacins, which were found to be δH 4.20–4.70/45.0–51.5 (C-5′), 172.0–172.9 (C-6′), and attributed to the amide linkage (Azumi et al. 2008; Li et al. 2012). The 3-(3′-methyl)-butyl isochromanone substructure was deduced by COSYs from H-1′, H-2′/H-3′/H-4′/H-5′/H-3 and HMBCs from H-4, H-5 to C-10; H-7 to C-8, C-9; H-1′, H-2′ to C-4′ and H-5′ to C-3, C-4, C-4′. Another substructure was constituted through COSYs from H-11′/H-10′/H-9′/H-8′, along with HMBCs from H-8′, H-9′ to C-7′; H-10′ to C-8′, C-11′, C-12′ and H-11′ to C-9′. The carbon shifts at δ 36.8 (C-10′) and 75.0 (C-8′) were characteristic of the attachment of nitrogen and oxygen to these carbons, respectively. The free $–NH_2$ was attached at C-10′ through HMBCs from δ 6.87 ($–NH_2$) to C-10′. The occurrence of hydroxyls at C-8 and C-8′ were further confirmed by deuterium exchange {deuterium oxide-^{1}H NMR} experiment. The previously reported lipoamicoumacins enclosed a tetrahydrofuranone with chemical shifts of δH 4.79/87.9 (C-9′), 4.51/48.6 (C-10′), 2.45/36.7 (C-11′), 178.1 (C-12′) [29], and these values were comparable with the titled compound {δH 4.47/81.3 (C-9′), 3.56/36.8 (C-10′), 3.15/68.4 (C-11′), 170.8 (C-12′)}. The data presented above along with previous reports contemplated for amicoumacin C-type structure, which were mostly isolated from *Bacillus* species (Li et al. 2015; Azumi et al. 2008). Other COSYs from H-15′ to H-18′, and HMBCs from H-15′ to C-14′, C-17′; H-16′, H-18′ to C-17′, and H-13′ to C-14′ attributed the butyl acetate side chain. The titled compound enclosed ten double bond equivalents corresponding to three double bonds, four carbonyls, and three rings. By comparison of NMR of previously reported amylomacin A (Azumi et al. 2008; Li et al. 2012, 2015), and through NOESYs, the relative conformations of protons were confirmed. NOEs from δH 3.07 (H-4β)/4.25 (H-5′)/4.32 (H-8′)/3.56 (H-10′)/3.15 (H-11′) deduced their equivalent plane of orientation, and hence, assigned as β-oriented. Similarly, NOEs between the protons at δH 4.47 (H-9′)/2.89 (H-4α)/4.57 (H-3) were recorded for their adjacent spatial orientation, and therefore, designated as α-oriented. The presence of IR band at 3435 cm^{-1} was found to be correlated with hydroxyl and the bands at 3232 (amino) and 1639 (amide) cm^{-1} could confirm the presence of two nitrogen functionalities in the titled compound. Following the nitrogen rule, the even nominal mass obtained from EI-MS spectra ($[M]^+$) and the odd nominal mass in the HRESIMS spectra ($[M+H]^+$) were indicative of the presence of even number of nitrogen functionalities. Previously reported amicoumacin derivatives were also supported for this nitrogen rule (Park et al. 2009). Herein, the nitrogen rule was well correlated with the titled compound as it exhibited *m/z* values of 520.2 $[M]^+$ and 521.2502 $[M+H]^+$ for EI-MS and HRESIMS analyses, respectively that confirmed the presence of two nitrogen groups in the structure. The molecular ion peak was obtained at *m/z* 520, and base peak was observed at *m/z* 147 (**I**; dihydroisocoumarin ion} (Li et al. 2012) through sequential fragmentations including McLafferty fragmentation.

Similar to amylomacin A, the structure of amylomacin B was confirmed through COSYs from H-5 to H-7; H-3/H-4; H-1′, H-2′/H-3′/H-4′/H-5′/H-3; H-11′/H-10′/H-9′/H-8′ together with HMBCs from H-3 to C-1, C-9; H-4 to C-10; H-5 to C-4, C-10; H-7 to C-8, C-9; H-4 to C-5′; H-5′ to C-3; H-8′, H-9′ to C-7′; H-10′ to C-8′, C-11′, C-12′ and H-11′ to C-9′. Presence of deshielded signals at δH 3.89/δC 71.2 in amylomacin B, when compared to the shielded signals at δH 1.92/δC 42.6 in amylomacin A ascribed the presence of hydroxyl attached methine in the former. The hydroxymethine at C-4′ was confirmed through HMBCs from H-1′, H-2′ to C-4′ and H-4′ to C-5′, C-3′. The proton at δ 7.65 was considered for –NH– adjacent to carbon at δ 45.5 (C-5′), and its position at H-6′ was ascribed through HMBCs from δ 7.65 to C-7′ and C-5′, H-5′ to C-7′. The $-NH_2$ was attached at C-10′ through HMBCs from δ 6.89 ($-NH_2$) to C-10′. Similar to the previous compound, the presence of hydroxyls at C-8 and C-8′ were confirmed by deuterium oxide-^{1}H NMR. When compared to previous compound, amylomacin B enclosed the methoxyethyl carboxylate side chain, which was ascribed by COSYs from H-14′/H-15′, and HMBCs from H-14′ to C-13′; H-15′ to C-16′, and H-16′ to C-15′. The methoxyethyl carboxylate was attached at C-11′ of amicoumacin B through HMBCs from H-10′, C-11′ to C-13′. Comparison of the NMR spectroscopic values of the previously reported compounds amylomacin A and B revealed that both shared the basic skeleton of dihydroisocoumarin-derived compound amicoumacin C. The amylomacin B enclosed ten double bond equivalents ascribed to three double bonds and three rings and four carbonyls. The studied compound exhibited similar relative sterochemistry as that of amylomacin A. NOEs from δH 3.10 (H-4β)/4.31 (H-5′)/4.50 (H-8′)/3.57 (H-10′)/3.14 (H-11′) could ascribe their β-orientation. Similarly, NOEs between the protons at δH 4.14 (H-9′)/3.89 (H-4′)/2.81 (H-4α)/4.60 (H-3) deduced their α-orientation. Similar to the previous compound, the presence of two nitrogen functionalities in amylomacin B was confirmed through the IR bands at 3225 (amino) and 1637 (amide) cm^{-1}. Amylomacin B reported *m/z* values of 524.2 $[M]^+$ and 525.2088 $[M+H]^+$ for EI-MS and HRESIMS analyses, respectively, which further confirmed the presence of two nitrogen groups as per the nitrogen rule. The mass fragmentations and base peak were found to be similar to those exhibited by amylomacin A.

Comparison of NMR values of amylomacin A and B revealed that both shared similar skeleton as that of amicoumacin C. Similar to amylomacin A, the compound amylomacin C was confirmed through COSYs from H-5 to H-7; H-3 to H-4; H-1′, H-2′/H-3′/H-4′/H-5′/H-3; H-11′/H-10′/H-9′/H-8′ and HMBCs from H-3 to C-1, C-9; H-4 to C-10; H-5 to C-10; H-6 to C-5; H-7 to C-8, C-9; H-1′, H-2′ to C-4′; H-5′ to C-4′; δ 7.63 (–NH–) to C-7′, C-5′; H-8′, H-9′, H-5′ to C-7′; H-10′ to C-8′, C-12′, H-11′ to C-9′; 6.83 ($-NH_2$) to C-10′. When compared to amylomacin A–B, the titled compound enclosed a butyl side chain, which was attributed through COSYs from H-13′/H-14′/H-15′/H-16′, and HMBCs from H-15′ to C-16′; H-16′ to C-14′; H-11′ to C-14′. Amylomacin C exhibited similar relative sterochemistry as that of amylomacin A-B. NOEs from δH 3.12 (H-4β)/4.33 (H-5′)/4.41 (H-8′)/3.41 (H-10′)/3.01 (H-11′) deduced the β-orientation of the protons. Similarly, NOEs between the protons at δH 4.51 (H-9′)/2.89 (H-4α)/4.60 (H-3) could attribute their α-orientation. Amylomacin C recorded *m/z* values of 462.2 $[M]^+$ and 463.2448 $[M+H]^+$ for EI-MS and HRESIMS analyses, respectively, which further confirmed the presence of two nitrogen groups in it (according to the nitrogen rule).

The studied amylomacins (A–C) exhibited antimicrobial potential against *P. aeruginosa*, VRE*fs*, *K. pneumoniae*, MRSA, *E. faecalis*, and *S. flexneri* with a range of minimum inhibitory concentration of 0.78–3.12 μg/mL, whereas ampicillin and chloramphenicol were active at 6.25–25 μg/mL (Table 2). Notably, the C-12′ amide group of amicoumcin could be crucial for antibacterial activity (Li et al. 2012). Topological polar surface area (tPSA) (designated as an electronic parameter) of the studied compounds were considerably superior (148–203) than that exhibited by chloramphenicol and ampicillin (115.4 and 109.5, respectively). This could also be analogized with the potential antagonistic properties of the studied amicoumacins isolated from the heterotrophic *B. amyloliquefaciens* over the standard antibiotics.

Bacillus amyloliquefaciens MTCC 12716 displayed specific amplification for functional *pks*/*nrps* gene bearing an amplicon of ≈ 1000 bp that was submitted in the NCBI GenBank (accession identity of MH157092). A previously reported study demonstrated that the amicoumacin-based isocoumarins were contemplated to biosynthesize by the hybrid *nrps* and *pks* gene clusters (Park et al. 2016). The titled amylomacin derivatives (A–C) were classified under the isocoumarin class of chemistry with the basic skeleton of 3,4-dihydro-8-hydroxy-isocoumarin, wherein the structural variations were due to the differences in the side chains at

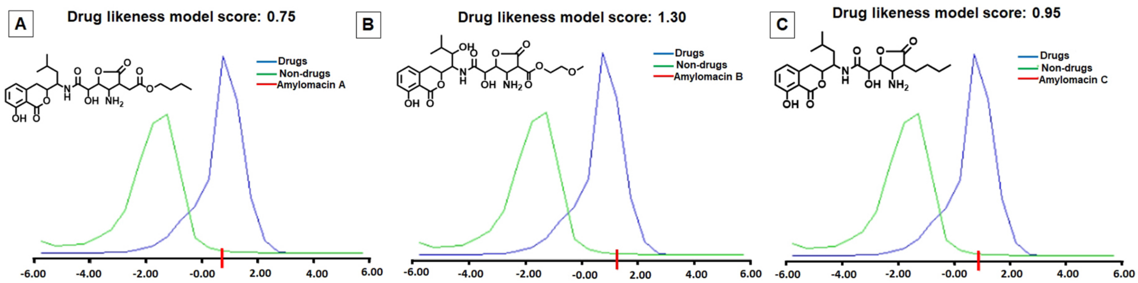


Fig. 4 **A**–**C** Drug-likeness score for amicoumacin derivatives (calculated with molsoft software)

the amide group as well as in the tetrahydrofuranone ring (Tang et al. 2016). The biosynthesis of amicoumacins includes the steps involving *N*-acyl-asparagine intermediates (such as preamicoumacins) that were hydrolyzed by the peptidase enzyme into an active constituent of amicoumacin. A putative scheme of biosynthetic pathway of amylomacins A–C by consecutive elongation modules through adenylations and condensations was elucidated (Fig. S37).

The drug-likeness model score for the titled amicoumacins (amylomacins A–C) were 0.75, 1.30 and 0.95, respectively. Conspicuously, the drug-likeness values for amylomacins A and C were on the lower side (0.75 and 0.95, respectively). Amylomacin B revealed a higher drug-like attribute (drug-likeness score of 1.30, Fig. 4), and therefore, could be a prospective antimicrobial lead attenuating the multidrug-resistant nosocomial pathogens.

Conclusions

Marine microorganisms are prospective resources to develop specialized antimicrobial compounds and drug leads necessitating renewable supply of unique bioactive molecules. In the quest to explore the bioactive metabolites of the intertidal seaweed-associated bacteria, we have investigated three homogenous amicoumacin derivatives, named as amylomacin A–C with undescribed structures and broad-spectrum antibacterial activities. These studied compounds were isolated to homogeneity from *B. amyloliquefaciens*, a heterotrophic bacterium associated with the seaweed. The amicoumacin compound encompassing 4′-hydroxy-11′-methoxyethyl carboxylate functionality (amylomacin B), displayed potential antagonistic activities against pathogenic bacteria, such as *P. aeruginosa*, VRE*fs*, *S. flexneri*, MRSA, *K. pneumoniae*, and *V. parahaemolyticus*. Amylomacin B displayed a higher drug-like attribute compared to other analogues. Putative biosynthetic pathway encompassing *pks*/*nrps* modular architecture was proposed, which reinforced the potential application of these class of compounds. From these results, amylomacin B warranted increasing consideration as a source of potential antibacterial agent for biotechnological and pharmaceutical applications.

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Author contributions The manuscript was written through contributions of all authors. KC conceived and designed research, acquired funds, conducted the experiments, analyzed data, and drafted the manuscript; VKK acquired funds and conducted experiments; KC, VKK, and MJ, and RDC analyzed data and drafted the manuscript. All authors read and approved the final version of the manuscript.

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Research involving human and/or animal participants This article does not contain any studies with human participants or animals performed by any of the authors.

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